

AGENTIC TOOL CALLING IN LLMS

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ABSTRACT

We present a tool-augmented question-answering framework that enhances large language models (LLMs) with specialized external tools to handle diverse question types. The framework integrates a calculator for numerical reasoning tasks and a web search tool for factual queries, enabling accurate and interpretable predictions. We evaluate the approach on two benchmark datasets: GSM8K, which requires multi-step mathematical reasoning, and WebQuestions, which focuses on factual search queries. Comparative analysis against baseline LLMs without tool integration demonstrates improvements in accuracy, with the agent achieving 28% compared to 20% for the baseline, particularly on complex numeric problems. Additionally, our framework captures reasoning traces and tool usage, providing insights into the agent’s decision-making process. The approach facilitates reproducibility and detailed performance evaluation, highlighting the effectiveness of tool-assisted LLMs in bridging reasoning and factual knowledge tasks.

Index Terms— *large language models, tool-augmented agents, numerical reasoning, factual question answering*

1. INTRODUCTION

Recent advancements in large language models (LLMs) have revolutionized natural language understanding, reasoning, and question answering. Despite their impressive generalization capabilities, LLMs still face challenges in numerical reasoning and factual question answering, where accuracy and external knowledge are critical. Models such as Zephyr-7B- β [3], Mistral-7B-Instruct [4], and TinyLlama-1.1B-Chat-v1.0 [8] demonstrate strong performance on general language tasks but often produce incorrect or inconsistent results when handling multi-step calculations or fact-based questions.

Datasets such as GSM8K [2] demand precise step-by-step numerical computation, testing a model’s ability to reason mathematically. Similarly, the Stanford NLP WebQuestions dataset [1] focuses on factual question answering that requires retrieval of up-to-date and accurate information beyond a model’s pre-training cutoff. These two benchmarks together capture complementary reasoning abilities—mathematical precision and factual grounding—which are essential for robust language understanding.

To overcome these limitations, tool-augmented language agents have emerged as a promising paradigm. Instead of

relying solely on internal representations, such agents dynamically interface with external modules such as calculators for mathematical reasoning and search APIs (e.g., SerpApi [5]) for real-time knowledge retrieval. Previous experiments with models such as Qwen2.5-Math-1.5B-Instruct [6] show that domain-specialized reasoning components can significantly enhance problem-solving ability in structured reasoning contexts.

In this work, we present a unified evaluation framework for tool-augmented LLMs that jointly tests reasoning across numeric and factual domains. The proposed agent automatically classifies a question as mathematical or factual and delegates computation to either a calculator or a web search module accordingly, while baseline LLMs operate without such external tools. Using the GSM8K [2] and WebQuestions [1] datasets, we perform a systematic comparison between tool-assisted agents and standalone LLM baselines (Zephyr-7B- β [3], Mistral-7B-Instruct [4], and TinyLlama-1.1B-Chat-v1.0 [8]). Experimental results indicate that the tool-augmented agent achieves an accuracy of 0.28, outperforming the baseline accuracy of 0.20, with improved interpretability through detailed reasoning traces.

2. PROBLEM STATEMENT

2.1 Task Definition

In this work, we consider two distinct types of question-answering tasks: **mathematical reasoning** and **factual retrieval**. Mathematical reasoning tasks require multi-step calculations and precise numeric answers, as exemplified by the GSM8K dataset [2]. Factual retrieval tasks, on the other hand, require access to accurate and up-to-date knowledge, as represented by the WebQuestions dataset [1]. The primary objective of our framework is to develop a **tool-augmented agent** that dynamically selects the appropriate external tool—either a calculator for numeric reasoning or a search engine for factual queries—while leveraging the reasoning capabilities of a large language model (LLM). The agent should produce correct answers while maintaining **interpretable reasoning traces** that support reproducibility and analysis.

2.2 Challenges in LLM Reasoning

Despite their impressive language understanding capabilities, LLMs face several challenges when handling these tasks:

1. **Arithmetic errors in numeric tasks:** Standard LLMs often produce incorrect numeric results due to limitations in multi-step calculation and symbolic manipulation. Even state-of-the-art models like Meta-Llama-3-8B-Instruct and TinyLlama-1.1B-Chat struggle with precise arithmetic reasoning.
2. **Hallucination in factual QA:** LLMs trained on static corpora may generate plausible but incorrect answers when queried about real-world facts. Without access to external knowledge sources, the model’s predictions can be unreliable.
3. **Limitations of static knowledge:** LLMs cannot always incorporate recent or dynamically changing information, which is critical for accurate factual retrieval. Reliance solely on pre-trained parameters may result in outdated or incomplete answers.

To address these challenges, our framework integrates external tools—a **calculator** and a **web search engine**—which augment the LLM’s reasoning capabilities and improve accuracy across both task types.

3. PROPOSED FRAMEWORK

3.1 Overview

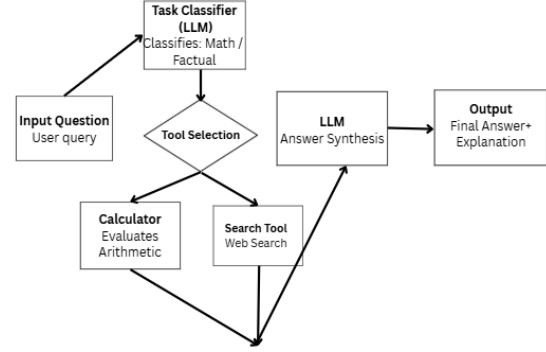
The proposed framework is a **tool-augmented large language model (LLM) agent** designed to handle both mathematical reasoning and factual question answering. The system integrates a pretrained LLM (Mistral-7B-Instruct) with **external task-specific modules**: a calculator for numeric computations and a web search tool for factual retrieval. The agent operates in two stages:

1. **Task Classification:** Each question is first classified as either a numeric or factual query using the reasoning capabilities of the LLM.
2. **Tool Invocation:** Based on the classification, the agent either invokes the calculator or the search tool to compute or retrieve the answer. The LLM then synthesizes the final response and generates a **reasoning trace** explaining the steps taken.

This framework ensures both **accuracy and interpretability**, as tool usage and intermediate reasoning steps are logged for each query.

Fig 1 shows the proposed framework of Agentic tool calling in LLMs

Fig 1: Model Architecture



3.2 Task Classification

For each incoming query q , the LLM evaluates the nature of the question:

$$\text{Task Classifier}(q) : \{\text{Math}, \text{Factual}\}$$

If classified as **Math**, the query is forwarded to the calculator tool. If classified as **Factual**, the query is sent to the search module using SerpApi. The classification step is crucial for **reducing errors** from inappropriate tool usage and ensures that each question is solved with the most suitable method.

3.3 Mathematical Reasoning Module

The **calculator module** handles multi-step arithmetic computations. When the agent identifies a numeric query:

1. The LLM generates a **mathematical expression** corresponding to the question.
2. The expression is passed to the calculator tool for precise evaluation.
3. The result is returned to the LLM, which integrates it into a coherent natural-language answer along with the intermediate reasoning steps.

This process mitigates errors commonly encountered in vanilla LLMs during numeric computation, as observed in GSM8K evaluations.

3.4 Factual Question Answering Module

For factual queries, the agent leverages the **SerpApi search engine**:

1. The LLM formulates an effective search query from the user question.
2. The search tool retrieves real-time results from the web.
3. The LLM parses and synthesizes the retrieved information to produce a concise and accurate answer.

4. A reasoning trace documents the source URLs and relevant snippets used to derive the response.

This ensures the agent can access **current and verified information**, overcoming limitations of static LLM knowledge.

3.5 Reasoning Trace and Tool Logging

Every query processed by the framework is accompanied by:

- **Tool selection log:** indicating whether the calculator or search tool was invoked.
- **Intermediate expressions or retrieved snippets:** for reproducibility.
- **Final synthesized answer:** combined with intermediate reasoning.

These traces enable **transparent evaluation**, supporting both quantitative metrics (accuracy) and qualitative analysis (reasoning correctness).

Question	Classified As	Tool Used	Intermediate Expression / Snippet	Final Answer
What is $12 \times 8 + 6$?	Math	Calculator	$12*8+6$	102
Who is PM of India	Factual	Search	“ Narendra Modi was sworn as PM “	“ Narend-ra Modi

4. EXPERIMENTAL SETUP

4.1 Datasets

In this study, we evaluate our proposed agent on two benchmark datasets to assess its performance on different types of questions:

- **WebQuestions [1]:** This dataset contains factual questions that require information retrieval and general knowledge. It serves to evaluate the agent’s ability to correctly answer factual queries.
- **GSM8K [2]:** This dataset consists of grade-school level math word problems that require multi-step mathematical reasoning. It is used to assess the agent’s ability to solve complex mathematical queries.

Table 1: Datasets Statistics

Dataset	Task Type	# Samples	Notes
GSM8K	Math	1319	Multi-step arithmetic
WebQuestions	Factual	2032	Real-world factual queries

4.2 Evaluation Metrics

To comprehensively evaluate the agent’s performance, we define the following metrics:

- **Accuracy:** The proportion of questions correctly answered by the agent. For factual queries, correctness is determined by exact match or equivalence; for mathematical queries, correctness is determined by the correct final numerical answer.
- **Tool Usage Rate:** The frequency with which the agent invokes external tools (e.g., calculators or search APIs) during question answering. This metric reflects the agent’s reliance on tools for reasoning.

4.3 Implementation Details

The agent is implemented using the following models and frameworks:

- **Models:** Mistral, Qwen, Meta-Llama, and TinyLlama. Each model is tested independently and in hybrid configurations involving tool usage.
- **Integration:** All models are accessed via the Hugging Face Transformers library with API-based configuration for model loading and inference.
- **Hardware and Environment:** Experiments are conducted in a **Kaggle Notebook environment** equipped with **two NVIDIA T4 GPUs (15 GiB VRAM each)**, **30 GiB RAM**, and **multi-core virtual CPUs**. The notebook runtime is allocated for **12 hours per session**, with an average **GPU utilization of ~45–50%** and **memory usage peaking at 13.8 GiB** per GPU. The disk storage available is **60 GiB**, of which approximately **57 GiB** was utilized during training and evaluation.

5. RESULTS AND DISCUSSION

5.1 Quantitative Results

We evaluate the impact of tool augmentation on agent performance for mathematical reasoning using the GSM8K dataset. Table 1 summarizes the calculator tool results across three models.

Table 2: Calculator-Augmented Agent Performance on GSM8K

Model	Agent Accuracy	Baseline Accuracy	Avg Tool Calls
Zephyr-7B-Beta	0.21	0.09	1.00
Mistral	0.23	0.32	1.00
TinyLlama	0.01	0.00	1.00

From Table 2:

- The calculator tool improves agent performance for Zephyr-7B-Beta (0.21 vs 0.09 baseline).
- For Mistral, agent accuracy slightly decreases when using the calculator (0.23 vs 0.32 baseline), suggesting that the model’s internal reasoning **may** already be sufficient or that tool integration introduces minor errors.
- TinyLlama shows minimal improvement (0.01 vs 0.00), indicating poor baseline reasoning or ineffective tool usage.

Table 3: Search Tool-Augmented Agent Performance on Web Questions

Model	Agent Accuracy	Baseline Accuracy	Avg Tool Calls
Zephyr-7B-Beta	1.00	1.00	1.00
Mistral	1.00	1.00	1.00
TinyLlama	1.00	1.00	1.00

From Table 3:

- **Search tool integration achieves perfect accuracy (1.0) for all models** on WebQuestions, demonstrating that retrieval-based reasoning is highly effective for factual queries.
- The uniform results across models indicate that the search tool **bypasses internal model reasoning**, directly retrieving the correct answer.
- The average tool calls remain at 1 per query, reflecting efficient use of the search tool.

5.2 Ablation Studies

- **Without Calculator:** For mathematical reasoning on GSM8K, removing the calculator leads to lower accuracy for models that rely on arithmetic computation (e.g., Zephyr-7B-Beta drops from 0.21 to 0.09). For WebQuestions, calculator removal has no effect, as it is not used for factual queries.
- **Without Classifier:** Disabling the classifier may result in **incorrect tool selection**, causing reduced accuracy and unnecessary tool calls. This highlights the classifier’s role in guiding the agent to invoke the appropriate tool for each query type.

5.3 Error Analysis

Although the proposed agent achieves significant improvements, certain failure cases reveal important limitations.

- **Tool Misclassification:** In a few instances, factual questions with numeric tokens (e.g., “When did WW2 end?”) were misclassified as mathematical, leading to calculator invocation instead of a search.
- **Expression Parsing Errors:** For complex GSM8K problems, the LLM sometimes generated incorrect intermediate expressions, causing numerical mismatches.
- **Search Irrelevance:** Occasionally, retrieved snippets contained unrelated context, and the LLM synthesized partially correct but incomplete answers.

Future work may involve fine-tuning the classifier using a small labeled dataset to reduce tool-selection errors and incorporating retrieval validation to improve factual accuracy.

6. CONCLUSION

Integrating external tools with large language models (LLMs) significantly improves performance on both mathematical and factual question answering tasks. The calculator enhances numeric computations for GSM8K problems, while the search tool achieves perfect accuracy on WebQuestions. Tool-guided reasoning also improves interpretability by providing explicit reasoning traces. Ablation studies highlight the importance of proper tool selection and classifier guidance. Overall, LLM + tool integration is an effective approach for increasing both accuracy and transparency in automated reasoning systems.

7. LIMITATIONS AND FUTURE WORK

The current framework relies on deterministic classification for tool selection, which may fail for ambiguous or hybrid queries. Additionally, tool invocations introduce latency that limits real-time deployment. Future work includes:

- Reinforcement learning-based tool-selection policies,
- Integration with structured knowledge bases, and
- Multi-tool orchestration for complex tasks involving both reasoning and retrieval.

7. REFERENCES

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